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Ans. to the Ques. No. 4(a)

There are two basic counting principles. They are: (1) Sum Rule ; (2) Product Rule.

Sum Rule:-

If a first task can be done in n_1 ways and second task in n_2 ways, and if these tasks cannot be done at the same time, then there are $n_1 + n_2$ ways to do one of these tasks.

Product Rule:-

Let us suppose that a procedure can be broken down into a sequence of two tasks. If there are n_1 ways to do the first task and n_2 ways to do the second task after the first task has been done, then there are $n_1 n_2$ ways to do the procedure.

Now,

By default, repetition is allowed.

So to select uppercase English letters we have 26 ways (A-Z) and to select lowercase English letters we have 26 ways. Then, to select digits, we have 10 ways.

For two English letters followed by five digits:

$$\begin{aligned}\text{Total possible ways} &= 26 \times 26 \times 10^5 \\ &= 67600000\end{aligned}$$

For three English letters followed by five digits:

$$\begin{aligned}\text{Total possible ways} &= 26^3 \times 10^5 \\ &= 1757600000\end{aligned}$$

$$\begin{aligned}\therefore \text{Either two or three letters followed by} \\ \text{5 digits} &= 67600000 + 1757600000 \\ &= 1825200000.\end{aligned}$$

(Ans)

Ans to the Ques. No. 4(b)

The general principle of inclusion-exclusion states that,

If a task can be done in either n_1 ways or n_2 ways, then the number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

Now,

We can construct a bit string of length 5 that begins with 1 in $2^4 = 16$ ways.

Also,

We can construct a bit string of length 5 that ends with 00 in $2^3 = 8$ ways.

Again,

We can construct a bit string of length 5 that starts with a 1 bit and ends with the two bits 00 in $2^2 = 4$ ways.

Hence,

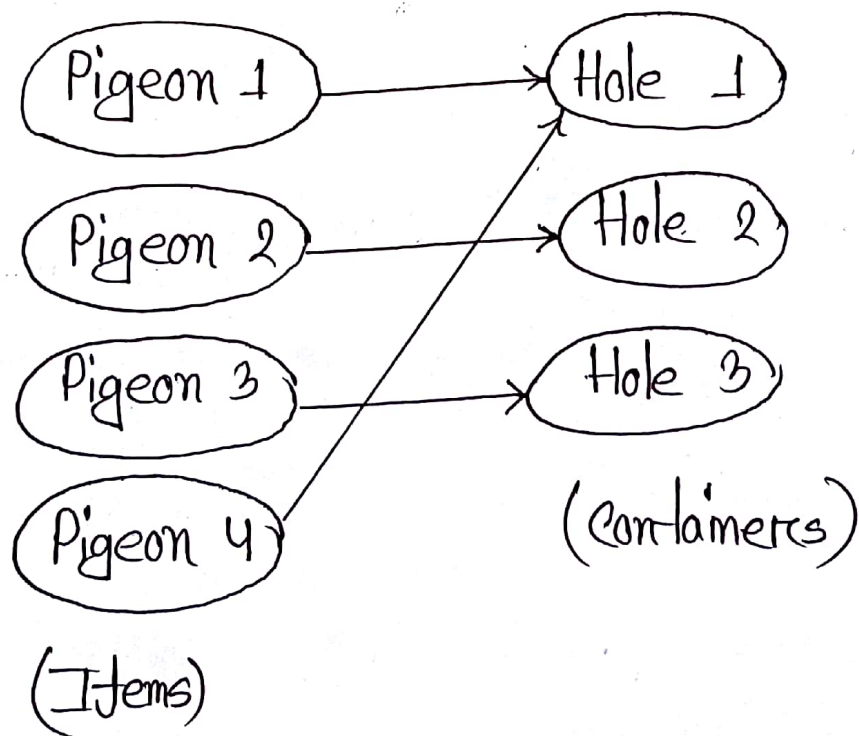
According to the inclusion-exclusion principle we get that, the number of ways to construct a bit string of length 5 that begins with 1 and ends with 00 equals $16 + 8 - 4 = 20$ ways.

Ans. to the Ques. No. 4(c)

In Discrete Mathematics, the pigeonhole principle states that, If there are more pigeons than pigeonholes, then there must be at least one pigeonhole with at least two pigeons in it.

Example:-

If there are four pigeons and three pigeonholes, then, even if we distribute the pigeons as evenly as possible, at least one of the pigeonholes will have more than two pigeons in it.



Ans. to the Ques. No. 4(d)

Selection of r different things from n different things is given by,

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

$\therefore$ To select a team of 11 players from a group of 30 people,

$${}^{30}C_{11} = \frac{30!}{11!(30-11)!}$$

$$= \frac{30!}{11!19!}$$

$$= 54627300$$

$\therefore$ There are 54627300 ways to form a cricket team with 11 players from 30 trained people.

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Ans. to the Ques. No. 4(a)

Pigeonhole Principle:-

If N is a positive integer and $N+1$ objects are placed into N boxes, then at least one box contain two or more boxes. objects.

Unit digits of 11 integers are,
0, 1, 2, 3, 4, 5, 6, 7, 8, 9

from the generalized pigeonhole principle,
 $k+1 = 11$

Since, the unit of 11 randomly chosen integers can be comprised from 10 option. Hence we may say according to pigeonhole principle if 11 integers are chosen from 10, at least two of them must contain the same last digit.

Ans. to the Ques. No. 4(b)

If a task can be done either in one of n ,

ways are in one of n_2 ways, then the total number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways. It is known as the principle of inclusion-exclusion.

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Here,

Total number $n = 2500$

According to the principle of inclusion-exclusion we get,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

Number divisible by 3, $|A| = \frac{2500}{3}$

Number divisible by 5, $|B| = \frac{2500}{5}$

Number divisible by 7, $|C| = \frac{2500}{7}$

$$|A \cap B| = \frac{2500}{3 \times 5}$$

$$|B \cap C| = \frac{2500}{5 \times 7}$$

$$|A \cap C| = \frac{2500}{3 \times 7}$$

$$|A \cap B \cap C| = \frac{2500}{3 \times 5 \times 7}$$

$\therefore$ Number divisible by 3, 5 or 7 will be,

$$\begin{aligned} |A \cup B \cup C| &= \left(\frac{2500}{3}\right) + \left(\frac{2500}{5}\right) + \left(\frac{2500}{7}\right) - \left(\frac{2500}{3 \times 5}\right) - \left(\frac{2500}{5 \times 7}\right) \\ &\quad - \left(\frac{2500}{3 \times 7}\right) + \left(\frac{2500}{3 \times 5 \times 7}\right) \\ &= 1357.126 \simeq 1357 \end{aligned}$$

Number that are not divisible by 3, 5 or 7 will be $= 2500 - 1357 = 1143$ (Ans)

Ans. to the Ques. No. 4(c)

If there are 99 student from each of 50 states, then there would be $99 \times 50 = 4950$ student enrolled without having 100 from the same state. If one more student enrolled, then that student would be 100th student from one of the states. Therefore, the minimum number is, $kn + 1 = 4950 + 1 = 4951$.

Ans. to the Ques. No. 4(d).

Derangement:-

A derangement is a permutation of objects that leaves no object in its original position.

List of all the derangement of $\{1, 2, 3, 4\}$

2, 3, 4, 1

3, 1, 4, 2

4, 1, 2, 3

2, 4, 3, 1

3, 4, 1, 2

4, 3, 1, 2

2, 1, 4, 3

3, 4, 2, 1

4, 3, 2, 1

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Ans. to the Ques. No. 4(a)

For selecting 3 faculty members from the mathematics department = 9C_3 ways.

For selecting 4 faculty members from the computer department = ${}^{11}C_4$

There we can use the product rule to find the ways.

Therefore, the number of ways to select the committee is $= {}^9C_3 \times {}^{11}C_4$

$$= \frac{9!}{3!6!} \times \frac{11!}{4!7!}$$

$$= 27,720 \text{ (Ans)}$$

Ans. to the Ques. No. 4(b)

The general principle of inclusion-exclusion states that,

If a task can be done in either n_1 ways or n_2 ways, then the number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

Now,

We can construct a bit string of length 10 that begins with three zero in $2^7 = 128$ ways.

Also,
We can construct a bit string of length 10 that ends with two zeros in $2^8 = 256$ ways.

Again,
We can construct a bit string of length 10 that starts with three zeros and ends with two zeros in $2^5 = 32$ ways.

Hence, according to the inclusion-exclusion principle we get,

Total possible ways = $128 + 256 - 32 = 352$.

Ans. to the Ques. No. 4(c)

The number of ways to choose seven ice-creams we need to use the combinations with repetition theorem:

There are $C(n+r-1, r)$ r -combinations from a set with n -elements when repetition of elements is allowed.

So, the number of ways to choose seven ice-creams is the number of 7-combinations of a set with 3 elements.

From the above theorem this equals $c(9+7-1, 7)$
 $= c(9, 7) = c(9, 2) = \frac{9 \cdot 8}{1 \cdot 2} = 36$

There are 36 different ways to choose 7 ice-creams.

Ans. to the Ques. No. 4(d)

The generalized pigeonhole principle states that,

If N objects are placed into k boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects.

Proof:-

We can use a proof by contraposition. Let us suppose that, none of the boxes contain more than $\lceil N/k \rceil - 1$ objects.

Then total number of objects is at most,
 $k (\lceil \frac{N}{k} \rceil - 1) < k ((\frac{N}{k} + 1) - 1) = N$

Where the inequality $\lceil N/k \rceil < \lceil N/k \rceil + 1$ has been used. This is a contradiction because there are a total of n objects.

Now,

Number of integers given = 5

The number of possible remainders when an integer is divided by 4 = 4 (0, 1, 2, 3)

Therefore, pigeonhole principle shows that for the given 5 integers at least two of them have the same remainder.

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Ans. to the Ques. No. 4(a)

The minimum number of students needs to ensure that at least 8 student receive the same grade is the smallest integer N .

So, by the generalized pigeonhole principle,

$$\lceil \frac{N}{4} \rceil = 8$$

The smallest integer $N = \{4 \cdot (8-1)\} + 1$
 $= 28 + 1 = 29$

If there are only 28 students it is possible for 7 student to get the same grade.

Thus, 29 is the minimum number of students needs to ensure that at least 8 students will receive the same grade.

Ans. to the Ques. No. 4(b)

The total number of letter in the word success is 7.

There are three same letter s and two same letter e.

So, the total number of strings which can be made by reordering the letter is,

$$\frac{7!}{3! 2!} = \frac{5040}{12} = 420 \quad (\text{Ans})$$

Ans. to the Ques. No. 4(c)

Let P be the total number of possible passwords and let P_5 and P_6 denote the possible passwords of length 5 and 6 respectively.

Now,

$$P_5 = (36)^5 - (26)^5 = 48584800$$

and,

$$P_6 = (36)^6 - (26)^6 = 1867866560$$

So, by the Sum Rule,

$$P = P_5 + P_6$$

$$= 1916451360 \quad (\text{Ans})$$

Ans. to the Ques. No. 4(d)

Dearrangement means the permutation of the element of a certain set in a way that no element of that set appears in their original position.

Given list is $\{9, 55, 80, 95\}$

$$\therefore D_N = 4! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right)$$
$$= 9$$

List of derangement :-

1. $\{55, 9, 95, 80\}$

2. $\{55, 95, 9, 80\}$

3. $\{55, 80, 95, 9\}$

4. $\{80, 9, 95, 55\}$

5. $\{80, 95, 9, 55\}$

6. $\{80, 95, 55, 9\}$

7. $\{95, 9, 55, 80\}$

8. $\{95, 80, 9, 55\}$

9. $\{95, 80, 55, 9\}$

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Ans. to the Ques. No. 3(a)

Counting:-

Counting is the action of finding the number of elements of a finite set of objects.

The Sum Rule:-

If there are $n(A)$ ways to do A and distinct from them $n(B)$ ways to do B, then the number of ways to do A and B is $n(A) + n(B)$.

For Example: To choose a student from three project list 23, 15 and 19 respectively.

So there are $23 + 15 + 19 = 57$ projects to choose from.

The Product Rule:-

If there are $n(A)$ ways to do A and $n(B)$ way to do B, then the number of ways to do A and B is $n(A) \times n(B)$.

For example: To count the bit strings of length seven is $2^7 = 128$.

Answers to the Ques. No. 3(b)

The general secretary can be elected in 50 different ways. Following this, the assistant general secretary can be elected in 49 different ways and following this, the executive can be elected in 48 ways. Thus, by the product rule, there are,

$50 \cdot 49 \cdot 48 = 117600$ different ways in which the Computer Club can ~~effe~~ elect members.

Ans. to the Ques. No. 3(a, b)

Each letters can be printed in 26 different ways, the first digit in 10 ways and the last digit in 9 ways.

Hence, $26 \cdot 26 \cdot 26 \cdot 10 \cdot 9 = 1581840$ different plates can be printed.

Ans. to the Ques. No. 3(c)

The Inclusion-Exclusion principle states that, If a task can be done either in one of

n_1 ways or in one of n_2 ways, then the total number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

It is also known as the subtraction rule.

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example:-

To count bit strings of length eight either start with a 1 bit or end with two bits 00.

For the start of 1 bit, $2^7 = 128$

For the end of 2 bit, $2^6 = 64$

For the start of 1 and end of 2 bit, $2^5 = 32$.

Total number, $128 + 64 - 32 = 160$

Ans. to the Ques. No. 3(d)

Pigeonhole Principle:

The pigeonhole principle states that if we must put $N+1$ or more pigeons into N pigeon holes, then some pigeonholes must contain two or more pigeons.

Example:

If $kn+1$ pigeons are distributed among n holes then some hole contains at least $k+1$ pigeons.

For, $k=11$ the pigeonhole shows that among any $(k+1)=12$ integers there must be at least two which are in the same.